



TASK 1

DSA



SEPTEMBER 13, 2026

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SU92-BSSEM-S25-052

PROBLEM 1

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int num{};
```

```
    cout << "Enter an Integer: ";
```

```
    cin >> num;
```

```
    int *ptr = &num;
```

```
    system("cls");
```

```
    cout << "Value of Num: " << num << endl;
```

```
    cout << "Address of Num: " << &num << endl;
```

```
    cout << "Value of Ptr: " << ptr << endl;
```

```
    cout << "Value of Num access through Ptr: " << *ptr << endl;
```

```
    return 0;
```

```
}
```

PROBLEM 2

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int num{};
```

```
cout << "Enter an Integer: ";

cin >> num;

int *ptr = &num;

cout << "Value Before Change: " << num << endl;

*ptr = 90;

cout << "Value After Change: " << num << endl;

return 0;

}
```

PROBLEM 3

```
#include <iostream>

using namespace std;

int main()

{

    int num1{}, num2;

    cout << "Enter an Two Integer: ";

    cin >> num1 >> num2;

    int *ptr1 = &num1, *ptr2 = &num2;

    system("cls");

    int sum, diff, product;

    sum = *ptr1 + *ptr2;
```

```
diff = *ptr1 - *ptr2;

product = *ptr1 * *ptr2;


cout << "Sum      : " << sum << endl;
cout << "Difference : " << diff << endl;
cout << "Product   : " << product << endl;


return 0;

}
```

PROBLEM 4

```
#include <iostream>
```

```
using namespace std;
```

```
void takeInput(int arr[], int size)
```

```
{
    cout << "Enter " << size << " Numbers: ";
    for (int i = 0; i < size; i++)
    {
        cin >> arr[i];
    }
}
```

```
void printArr(int arr[], int size)
```

```
{
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
}
```

```

    }

    cout << endl;
}

void printRequirdElement(int arr[], int size)
{
    int smallest = INT_MAX;
    int largest = INT_MIN;

    for (int i = 0; i < size; i++)
    {
        if (smallest > arr[i])
            smallest = arr[i];

        if (largest < arr[i])
            largest = arr[i];
    }

    cout << "Largest: " << largest << endl;
    cout << "Smallest: " << smallest << endl;
}

int main()
{
    int size = 5;
    int arr[size];

    takeInput(arr, size);
    system("cls");
}

```

```
    printArr(arr, size);  
    printRequirdElement(arr, size);  
}
```

PROBLEM 5

```
#include <iostream>
```

```
using namespace std;
```

```
void takeInput(int arr[], int size)  
{  
    cout << "Enter " << size << " Numbers: ";  
    for (int i = 0; i < size; i++)  
    {  
        cin >> arr[i];  
    }  
}
```

```
void printRequirement(int arr[], int size)  
{  
    int evenCount{}, oddCount{};  
  
    for (int i = 0; i < size; i++)  
    {  
        if (arr[i] % 2 == 0)  
            evenCount++;  
  
        else  
            oddCount++;  
    }
```

```
    cout << "Even Count: " << evenCount << endl;  
    cout << "Odd Count: " << evenCount << endl;  
}
```

```
int main()  
{  
    int size = 10;  
    int arr[size];  
  
    takeInput(arr, size);  
    system("cls");  
    printRequirement(arr, size);  
}
```